

Close the loop close to APD pins

Place R_{ISO} close to IN_-

Thermal Pad

1 PB 8 V_{S+}

2 FB 7 OUT

3 IN_- 6 NC

4 IN_+ 5 V_{S-}

10uW optical
 -> 4.4 uA photocurrent
 -> $4.4\mu\text{A} \cdot 15\text{k}\Omega = 66\text{mV}$

float or ground
 the diode can

C126
 1pF: install as

OPA818x1
 U1

Farnell 1495575
Hamamatsu S5973
D= 400um active area
Cd=1.6 pF @ Vr=3V3

Thorlabs FDS015
D=150 μ m active area
0.65pF @ $V_r = 5$ V

Thorlabs FDS025
D= 250 μm active area, with ball lens
0.94 pF @ $V_r = 5 \text{ V}$

Hamamatsu S5972
D= 800um active area
3 pF @ $V_r = 10\text{ V}$

Hamamatsu S5971
D= 1.2 mm active area
3 pF @ $V_r = 10\text{ V}$

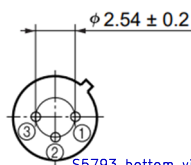
OSI FCI-125G-006HRL
Hamamatsu S9055(-01)

Thorlabs FGA01FC, InGaAs, 2pF @ 5V
same as OSI INGAAS-120L-FC

Thorlabs FGA015, InGaAs, 1.5 pF @ 5V
D = 0.15 mm active area

FGA01, ball lens

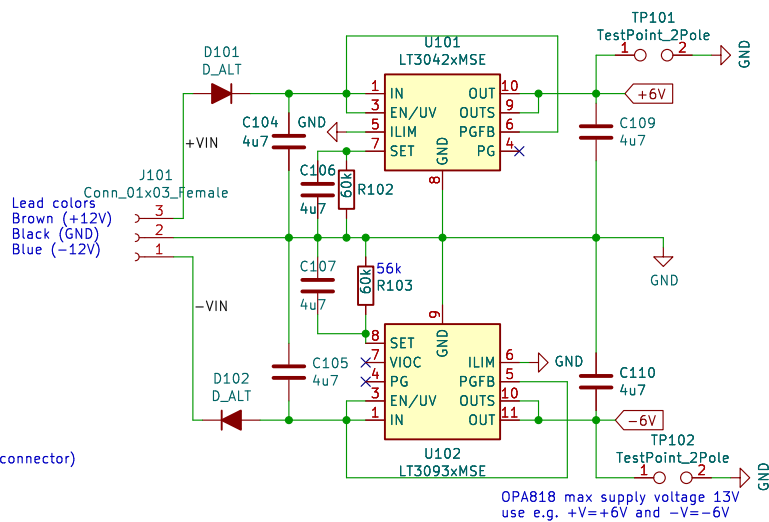
For estimating Bandwidth, Transimpedance–amplifier noise, shot–noise limit, etc., see: TIASim at <https://github.com/aewallin/TIASim>



S5793 bottom view



CASE

$$R_{set} = V_{out} / 100\mu A$$


PSU:
Thorlabs LDS12B
±12 VDC Regulated Linear Power Supply, 6 W
Lumberg RSMV3 male connector, 3-pin

PSU-lead: Digikey A120947-ND (mates with Thorlabs 3-pin psu connector)
3-pin M8 female connector
1.5 m 3-wire lead, solder to PCB

See MeasurementData sheet for measurements and noise-models for built 2-stage detectors:

MeasurementData

File: 2stageTIA_measurementData.kicad_sch

2019 detector (OPA657) gain:
78 dBohms transimpedance (8k2 ohms)
+20 dB postgain (10V/V)
+28 dB postgain (ZFL-500LN)
126 dB total

2019Modified detector (OPA657) gain:
92 dBohms transimpedance (39k ohms)
+20 dB postgain (10V/V)

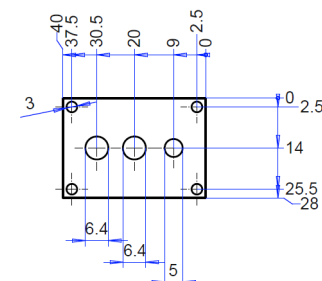
112 dB total

2022 detector with OPA818
90 dBohms transimpedance (33kOhm)
20 dB postgain (10 V/V)
110 dB total

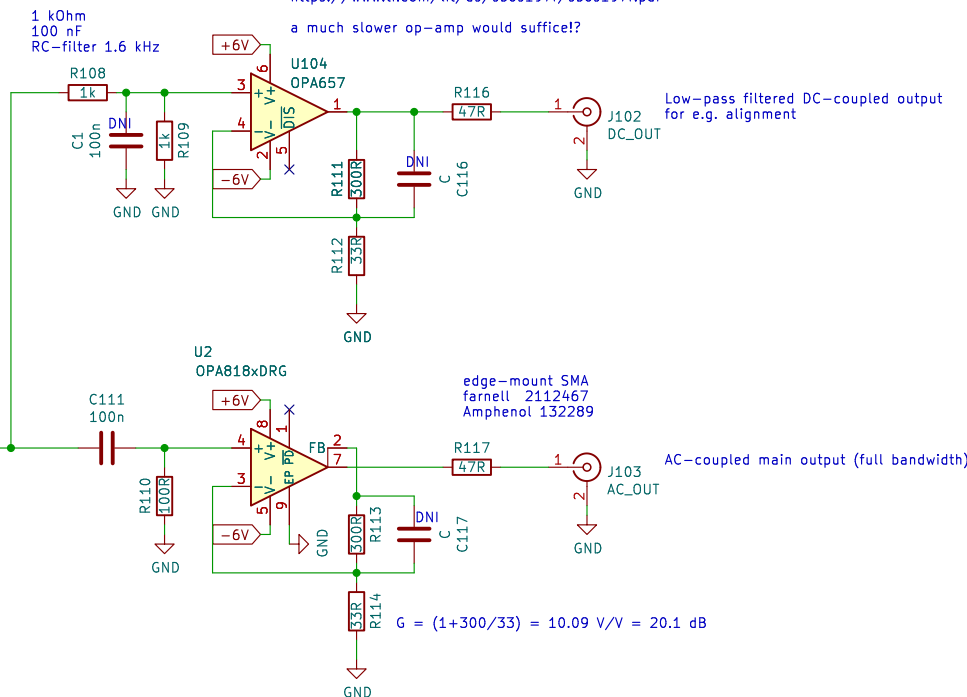
Mechanical design for enclosure top



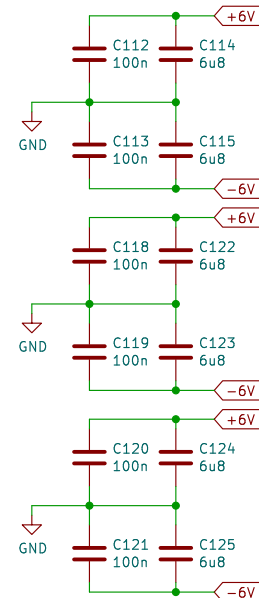
D6.4 mm holes for SMA-connectors
D5 mm hole for power-supply cables



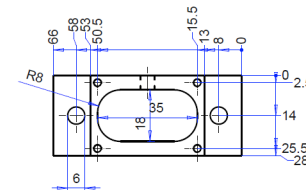
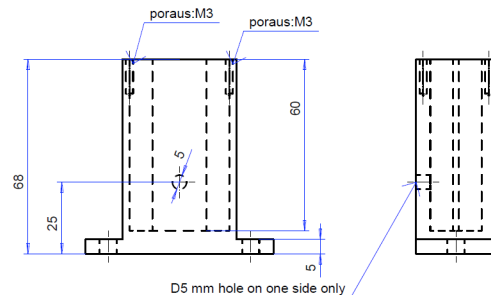
use OPA657 here?
<https://www.ti.com/lit/ds/sbos197f/sbos197f.pdf>
 a much slower op-amp would suffice!?


$$G = (1 + 300/33) = 10.09 \text{ V/V} = 20.1 \text{ dB}$$

Decoupling caps for three op-amps



Mechanical design of Aluminium enclosure
Optical beam—height over table 25mm
Mounting—holes for M6 screws on 25mm grid



2022-10 built with FGA015 InGaAs photodiode and $R_f=30k\Omega$, ca 44 MHz BW
 2022-06: built 1st board with $R_f=33k\Omega$, 10 V/V postgain – measurements look good.
anders.e.e.wallin "at" gmail.com

Sheet: /
File: working.kicad_sch

Title: 2-stage OPA818 photodetector

Size: A3	Date: 2019-12
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KiCad E.D.A.	eeschema 7.0.6
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Rev: draft

Id: 1/2

1	2	3	4	5	6
A					A
B					B
C					C
D					D
1	2	3	4	5	6

Sheet: /MeasurementData/ File: 2stageTIA_measurementData.kicad_sch		
Title:		
Size: A4	Date:	Rev:
KiCad E.D.A. eeschema 7.0.6	Id: 2/2	